

TRENTON-MERCER, NJ, USA

WMO: 724095

Lat: 40.277N

Lon: 74.816W

Elev: 58

StdP: 100.63

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14792

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-9.1	-21.1	0.6	-9.9	-18.5	0.7	-7.5	10.6	1.7	9.3	1.1	3.7	320	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.5	23.5	32.1	22.9	30.7	22.4	25.1	30.8	24.4	29.3	23.8	28.2	3.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.5	27.5	22.9	17.7	26.5	22.4	17.2	26.2	76.7	30.8	73.8	29.4	71.2	28.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	8.0	7.2	-14.1	35.7	2.7	2.1	-16.0	37.2	-17.5	38.4	-19.0	39.6	-20.9	41.2		
			-15.2	26.6	2.4	0.8	-16.9	27.2	-18.4	27.6	-19.7	28.1	-21.5	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	0.0	1.4	5.6	11.6	17.0	21.9	24.7	23.8	20.0	13.5	7.9	3.0
	DBStd	9.62	5.61	5.08	5.11	4.64	4.20	3.56	2.84	2.84	3.63	4.44	4.51	5.05
	HDD10.0	1079	313	244	155	33	1	0	0	0	0	16	93	225
	HDD18.3	2707	569	473	394	208	77	11	0	1	24	161	313	476
	CDD10.0	2025	2	3	19	81	219	356	456	428	299	124	29	8
	CDD18.3	613	0	0	1	7	37	116	198	171	73	11	1	0
	CDH23.3	5359	0	0	6	78	366	1023	1868	1430	521	68	1	0
	CDH26.7	1880	0	0	0	18	107	347	751	502	144	11	0	0
Wind	WSAvg	3.1	3.4	3.5	3.8	3.7	3.0	2.8	2.6	2.4	2.7	2.9	3.1	3.2
Precipitation	PrecAvg	1074	79	75	91	79	87	87	125	100	106	72	79	94
	PrecMax	1509	228	131	186	168	158	159	340	245	230	138	197	177
	PrecMin	772	9	25	24	21	6	16	33	12	20	1	8	15
	PrecStd	193	47	31	39	35	43	45	81	62	58	36	50	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	19.2	23.6	29.6	31.8	34.0	36.4	35.0	33.1	27.9	22.1	19.5
		MCWB	14.4	13.7	15.5	19.1	21.8	23.8	23.7	24.1	23.0	20.3	16.9	15.9
	2%	DB	14.1	15.3	19.5	26.0	29.7	32.1	33.9	32.8	30.2	25.2	19.5	16.3
		MCWB	11.5	11.2	13.2	16.6	20.7	22.9	23.6	23.5	22.1	19.7	15.2	13.8
	5%	DB	10.6	11.9	16.7	22.7	27.4	30.3	32.3	31.1	28.0	22.8	17.4	13.2
		MCWB	7.8	7.7	11.1	15.1	19.6	22.2	23.1	22.6	21.0	18.1	13.5	10.9
	10%	DB	7.8	8.8	13.9	20.0	25.1	28.6	30.8	29.4	26.3	20.9	15.3	10.7
		MCWB	5.3	5.2	9.1	13.3	18.4	21.1	22.6	21.8	20.1	16.7	12.1	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	15.6	16.7	20.5	23.1	25.1	26.3	26.2	24.8	22.5	18.4	16.8
		MCDB	16.8	18.4	22.0	26.1	29.3	31.2	33.2	32.1	29.7	25.4	20.4	18.6
	2%	WB	12.0	11.9	14.4	18.4	21.8	24.2	25.3	25.1	23.7	20.8	16.6	14.1
		MCDB	13.5	14.8	18.3	23.7	27.7	29.5	31.4	30.1	27.4	24.0	18.8	15.7
	5%	WB	8.1	7.9	11.9	16.5	20.7	23.5	24.5	24.2	22.8	19.3	14.6	11.5
		MCDB	10.0	10.7	15.4	21.2	25.9	28.4	29.8	28.5	26.0	22.1	16.7	13.1
	10%	WB	5.6	5.6	9.5	14.7	19.5	22.5	23.8	23.5	21.8	17.5	12.7	8.4
		MCDB	7.6	8.5	13.6	18.9	23.6	27.0	28.5	27.3	24.9	19.9	14.7	10.1
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	9.6	10.8	10.6	10.1	10.1	9.7	10.1	9.8	9.4	7.8
		MCDBR	11.7	13.1	14.0	14.8	13.5	11.8	11.8	11.2	11.8	11.7	11.7	10.7
		MCWBR	10.0	9.9	9.1	7.9	6.5	5.1	4.3	4.2	5.4	6.7	8.2	9.6
	5% WB	MCDBR	11.1	12.2	12.7	12.7	11.8	10.5	10.0	9.3	9.4	9.6	9.9	10.1
		MCWBR	10.1	10.3	9.6	8.1	6.2	5.2	4.3	4.2	5.2	6.7	8.2	10.0
Clear-Sky Solar Irradiance	taub	0.308	0.320	0.358	0.387	0.433	0.496	0.492	0.485	0.400	0.358	0.327	0.307	
	taud	2.464	2.421	2.381	2.330	2.227	2.093	2.119	2.125	2.338	2.462	2.511	2.521	
	Ebn at Noon	856	896	894	887	849	794	792	785	838	841	828	832	
	Edh at Noon	78	94	109	123	139	160	154	149	113	90	75	69	
All-Sky Solar Radiation	RadAvg	1.90	2.81	3.71	4.80	5.32	5.84	5.97	5.14	4.29	3.04	2.15	1.64	
	RadStd	0.17	0.30	0.30	0.44	0.61	0.50	0.29	0.33	0.47	0.31	0.19	0.17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (68 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page